
SUMMARY

Mechanical Engineering MS student at the University of Washington proficient in high-fidelity CFD simulation (STAR-CCM+, ANSYS Fluent), turbulence modeling and designing. Proven track record across aerospace, biomedical, and energy applications — from turbine cascade aerodynamics to patient-specific airway flows. Combines deep simulation fundamentals with emerging machine learning methods (PINNs) and strong programming skills to automate workflows and extract actionable engineering insight.

SKILLS

Simulation & Analysis: ANSYS Fluent, STAR-CCM+, COMSOL, Ansys Workbench, OriginPro, TecPlot, FEA

Turbulence Modeling: RANS (k- ω SST), LES, PANS, DNS-level reconstruction via PINNs

CAD & Design: SolidWorks, Fusion 360, SolidEdge, AutoCAD, OpenRocket

Programming: Python, MATLAB, C++, Arduino IDE, TUI/UDF scripting (ANSYS Fluent)

Domain Expertise: Internal/external aerodynamics, multiphase flow, thermal-fluid coupling, biomedical flows.

EXPERIENCE

Multiphase and cardiovascular flow dynamics | Graduate Student Researcher, UW, Seattle

Jan 2026 - ongoing

- Simulate complex internal flow dynamics in patient-specific airway geometries (tracheomalacia) using STAR-CCM+ and in-house Python codes, improving informed clinical treatment planning by predictive 43% .
- Conduct experimental flow analysis of the left ventricle with an Impella assist device, characterizing hemodynamic performance to support cardiac device optimization.

Improvements and Innovation in Particle Tracking Velocimetry (PTV) | Graduate Student Researcher, UW, Seattle

Sep 2025 - ongoing

- Developing Physics-Informed Neural Network (PINN) architectures to reconstruct full 2D/3D flow fields from sparse experimental PTV data, targeting DNS-level accuracy, ensuring improvement in data obtained from measurements.

Grader (ME 355: Introduction to Manufacturing Processes) | Department of Mechanical Engineering, UW Seattle

Jan 2026 - ongoing

- Evaluate student work across manufacturing fundamentals; maintain active knowledge of manufacturing techniques.

Numerical Prediction and Separation control | Summer Research Intern, IIT Kanpur, India

May 2024 - Aug 2024

- Modelled and analyzed external flow separation on low-pressure turbine cascade geometries using RANS and LES turbulence models in ANSYS Fluent on HPC, achieving 75% improvement in computational efficiency.
- Evaluated passive flow control strategies to recover blade efficiency .
- Presented findings at The Asian Congress on Gas Turbines (ACGT) 2024.

Undergraduate Student Researcher | IIT Bhilai, India

Sep 2023 - May 2025

- Transdermal drug delivery: Developed in-house MATLAB code to model and optimize non-invasive drug diffusion through skin; used COMSOL to simulate porous medium flow through a custom Stratum Corneum geometry.
- Building energy: Built a MATLAB-based Natural Ventilation Potential(NVP) Hours predictor, reducing modeled HVAC building energy consumption by 20–26% through passive thermal system design.

CFD Investigation of Solidification during Single Crystal Growth and Transitional Turbulence Using PANS Models

Summer Intern, Defense Research and Development Organization (DRDO) India

May 2023- July 2023

- Simulated thermal-fluid coupling in metallic solidification using ANSYS Fluent with custom User Defined Functions (UDFs).
- Applied Partially Averaged Navier–Stokes (PANS) turbulence models to capture transitional flow regimes.
- Implemented TUI scripting and solver-level modifications(C++) for problem-specific modeling and control.

EDUCATION

M.S. in Mechanical Engineering

Expected June 2027

University of Washington, Seattle, WA | GPA: 3.91/4.0

Relevant Coursework: Control Theory, Compressible Fluid Mechanics, Structural Composites in Aircraft(Boeing)

B.Tech (Honors) in Mechanical Engineering

Nov 2021 – May 2025

Indian Institute of Technology Bhilai, India | GPA: 9.0/10.0

Relevant Coursework: Computational Fluid Dynamics, Combustion in IC Engines, Additive Manufacturing

Activities: Braking Department, Motorsports Team (eBaja, mBaja, Formula); Social Executive, National Service Scheme